CSA4301-SLOT D

INTERNET PROGRAMMING

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EXP 1

Aim:

To design and develop a College Library Management System website using HTML, CSS, JavaScript, Node.js, Express, and MySQL to automate library operations, including book searching, student registration and login, book issue and return management, fine calculation, inventory maintenance, and year-wise and semester-wise study material uploads and downloads, providing an efficient and user-friendly platform for students and librarians.

HTML CODE:

```
<!DOCTYPE html>
<html lang="en">
<head>
<meta charset="UTF-8">
<meta name="viewport" content="width=device-width, initial-scale=1.0">
<title>College Library Management System</title>

<style>
:root {
  --navy: #142f50;
  --blue: #2563eb;
  --light: #f4f7fc;
  --text: #26364b;
  --muted: #718096;
  --border: #e5eaf2;
  --green: #15803d;
  --red: #dc2626;
}
* { box-sizing: border-box; }
body {
  margin: 0;
  font-family: Arial, sans-serif;
  background: var(--light);
  color: var(--text);
}
button, input, select, textarea { font: inherit; }
```

```
button { cursor: pointer; }
button:focus-visible, a:focus-visible, input:focus-visible {
    outline: 2px solid #60a5fa;
    outline-offset: 2px;
}
.header {
    background: var(--navy);
    color: white;
    text-align: center;
    padding: 24px 16px 18px;
}
.brand {
    display: flex;
    align-items: center;
    justify-content: center;
    gap: 14px;
}
.brand-icon { font-size: 42px; }
.brand h1 { font-size: clamp(21px, 3vw, 31px); margin: 0; }
.header p { margin: 10px 0 0; color: #e2eaf6; }
.navbar {
    background: #203f64;
    display: flex;
    justify-content: center;
    flex-wrap: wrap;
    gap: 5px;
    padding: 9px;
}
.navbar button {
    background: transparent;
    color: white;
    border: 0;
    border-bottom: 3px solid transparent;
    padding: 12px 15px;
}
.navbar button:hover, .navbar button.active {
    border-bottom-color: #60a5fa;
    background: #ffffff0d;
}
```

```
main {
  width: min(1200px, 95%);
  margin: 25px auto 45px;
}
.page { display: none; }
.page.active { display: block; }
.hero {
  min-height: 275px;
  padding: 36px;
  border-radius: 16px;
  background:
    linear-gradient(90deg, #ffffffef 0%, #ffffffd9 50%, #ffffff25 100%),
    url("https://images.unsplash.com/photo-1507842217343-583bb7270b66?auto=format&fit=crop&w=1600&q=85")
    center/cover;
  display: flex;
  flex-direction: column;
  justify-content: center;
}
.hero h2 { font-size: clamp(30px, 4vw, 43px); margin: 0 0 12px; color: var(--navy); }
.hero p { max-width: 460px; line-height: 1.7; }
.search-box { display: flex; gap: 8px; max-width: 570px; margin-top: 14px; }
input, select, textarea {
  width: 100%;
  border: 1px solid #d6deea;
  border-radius: 7px;
  padding: 12px;
  background: white;
  color: var(--text);
}
.search-box input { flex: 1; min-width: 0; }
.btn {
  border: 0;
  border-radius: 7px;
  padding: 11px 16px;
  background: var(--blue);
  color: white;
}
.btn:hover { background: #1d4ed8; }
.btn.secondary { background: #e8effa; color: var(--navy); }
```

```
.btn.danger { background: var(--red); }
.btn.success { background: var(--green); }
.btn.small { padding: 7px 10px; font-size: 13px; }
.section-head {
  display: flex;
  justify-content: space-between;
  align-items: center;
  flex-wrap: wrap;
  gap: 12px;
  margin: 28px 0 16px;
}
.section-head h2, .section-head h3 { margin: 0; }
.muted { color: var(--muted); }
.features {
  display: grid;
  grid-template-columns: repeat(4, minmax(0, 1fr));
  gap: 16px;
  margin: 24px 0;
}
.feature {
  background: white;
  border: 1px solid var(--border);
  border-radius: 12px;
  padding: 22px 12px;
  text-align: center;
  box-shadow: 0 3px 12px #142f5008;
}
.feature .emoji { font-size: 28px; }
.feature h3 { font-size: 16px; margin: 12px 0 8px; }
.feature p { font-size: 13px; line-height: 1.5; color: var(--muted); }
.feature button { margin-top: 5px; }
.book-grid {
  display: grid;
  grid-template-columns: repeat(3, minmax(0, 1fr));
  gap: 16px;
}
.book-card {
  display: flex;
  gap: 13px;
```

```
    padding: 15px;
    background: white;
    border: 1px solid var(--border);
    border-radius: 11px;
    min-width: 0;
}
.book-cover {
    width: 76px;
    min-width: 76px;
    height: 108px;
    display: grid;
    place-items: center;
    text-align: center;
    padding: 7px;
    color: white;
    background: #24476b;
    border-radius: 4px;
    font-size: 11px;
    font-weight: bold;
    overflow-wrap: anywhere;
}
.book-info { min-width: 0; }
.book-info h3 { font-size: 15px; margin: 2px 0 8px; }
.book-info p { font-size: 12px; color: var(--muted); }
.book-info .btn { font-size: 12px; padding: 8px 10px; }
.panel {
    background: white;
    border: 1px solid var(--border);
    border-radius: 12px;
    padding: 22px;
    margin-bottom: 20px;
}
.panel h2, .panel h3 { margin-top: 0; }
.stats {
    display: grid;
    grid-template-columns: repeat(4, minmax(0, 1fr));
    gap: 15px;
    margin-bottom: 22px;
}
```

```
.stat {
  background: white;
  border: 1px solid var(--border);
  border-radius: 11px;
  padding: 19px;
}
.stat p { color: var(--muted); margin: 0 0 9px; font-size: 13px; }
.stat strong { font-size: 28px; color: var(--navy); }
.form-grid {
  display: grid;
  grid-template-columns: repeat(2, minmax(0, 1fr));
  gap: 14px;
}
.field { display: flex; flex-direction: column; gap: 7px; }
.field.full { grid-column: 1 / -1; }
.field label { font-size: 13px; font-weight: bold; }
.form-actions { margin-top: 16px; display: flex; gap: 9px; flex-wrap: wrap; }
.table-wrap { overflow-x: auto; }
table { width: 100%; border-collapse: collapse; font-size: 13px; }
th, td { padding: 13px 10px; text-align: left; border-bottom: 1px solid var(--border); }
th { background: #f5f8fc; color: #475569; white-space: nowrap; }
td { vertical-align: middle; }
.badge {
  display: inline-block;
  padding: 5px 8px;
  border-radius: 20px;
  background: #dcfce7;
  color: #166534;
  font-size: 11px;
}
.badge.unavailable { background: #fee2e2; color: #991b1b; }
.toolbar { display: flex; gap: 10px; flex-wrap: wrap; margin-bottom: 16px; }
.toolbar > * { flex: 1; min-width: 160px; }
.material-grid {
  display: grid;
  grid-template-columns: repeat(3, minmax(0, 1fr));
  gap: 14px;
}
.material {
```

```
    border: 1px solid var(--border);
    border-radius: 10px;
    padding: 18px;
}
.material h3 { font-size: 16px; }
.message { margin-top: 12px; font-size: 14px; }
.notice { background: #eff6ff; padding: 13px; border-radius: 8px; line-height: 1.5; font-size: 13px; }
footer {
    background: var(--navy);
    color: white;
    padding: 25px;
    text-align: center;
}
footer p { margin: 5px; font-size: 13px; }
.empty { padding: 24px; text-align: center; color: var(--muted); }
[hidden] { display: none !important; }

@media (max-width: 850px) {
    .features { grid-template-columns: repeat(2, 1fr); }
    .book-grid { grid-template-columns: repeat(2, 1fr); }
    .stats { grid-template-columns: repeat(2, 1fr); }
    .material-grid { grid-template-columns: repeat(2, 1fr); }
}
@media (max-width: 520px) {
    .brand { flex-direction: column; gap: 5px; }
    .navbar button { padding: 10px; font-size: 12px; }
    .hero { padding: 25px 18px; }
    .features, .book-grid, .stats, .material-grid, .form-grid {
        grid-template-columns: 1fr;
    }
    .field.full { grid-column: auto; }
    .search-box { flex-direction: column; }
    .panel { padding: 16px; }
}
</style>
</head>

<body>
<header class="header">
```

```
    <div class="brand">
      <span class="brand-icon">📖</span>
      <h1>College Library Management System</h1>
    </div>
    <p>Your digital library, all in one place.</p>
</header>

<nav class="navbar" aria-label="Main navigation">
    <button class="active" data-page="home">Home</button>
    <button data-page="books">Books</button>
    <button data-page="materials">Study Materials</button>
    <button data-page="login">Login</button>
    <button data-page="register">Register</button>
    <button data-page="contact">Contact</button>
    <button data-page="admin">Librarian</button>
    <button data-page="profile">My Profile</button>
</nav>

<main>
<!-- HOME -->
<section class="page active" id="home">
    <div class="hero">
      <h2>Welcome to<br>Our Library</h2>
      <p>Discover books, learning materials, and knowledge for a better future.</p>
      <div class="search-box">
        <input id="homeSearch" placeholder="Enter book title or author">
        <button class="btn" onclick="searchFromHome()">Search</button>
      </div>
      <p id="homeSearchMessage" role="status"></p>
    </div>

    <div class="features">
      <article class="feature">
        <div class="emoji">▧ </div>
        <h3>Search Books</h3>
        <p>Find your favorite books easily.</p>
        <button class="btn small" onclick="showPage('books')">Explore Books</button>
      </article>
      <article class="feature">
```

```
        <div class="emoji">👤</div>
        <h3>User Accounts</h3>
        <p>Register and manage your library account.</p>
        <button class="btn small" onclick="showPage('login')">Login</button>
      </article>
      <article class="feature">
        <div class="emoji">📄</div>
        <h3>Study Materials</h3>
        <p>Access year-wise and semester-wise materials.</p>
        <button class="btn small" onclick="showPage('materials')">View Materials</button>
      </article>
      <article class="feature">
        <div class="emoji">🎧</div>
        <h3>Contact Us</h3>
        <p>Get in touch with the librarian.</p>
        <button class="btn small" onclick="showPage('contact')">Contact</button>
      </article>
    </div>

    <div class="section-head">
      <h2>Featured Books</h2>
      <button class="btn secondary" onclick="showPage('books')">View All →</button>
    </div>
    <div class="book-grid" id="featuredBooks"></div>
  </section>

  <!-- BOOK CATALOG -->
  <section class="page" id="books">
    <div class="section-head">
      <h2>Book Catalog</h2>
      <span class="muted">Search by title, author or category</span>
    </div>
    <div class="panel">
      <div class="toolbar">
        <input id="catalogSearch" placeholder="Search books..." oninput="renderBooks()">
        <select id="categoryFilter" onchange="renderBooks()">
          <option value="">All Categories</option>
          <option>Computer Science</option>
          <option>Programming</option>
```

```
            <option>DBMS</option>
            <option>Networking</option>
            <option>General</option>
          </select>
        </div>
        <div class="table-wrap">
          <table>
            <thead><tr><th>Book</th><th>Author</th><th>Category</th><th>Availability</th><th>Action</th></tr></thead>
            <tbody id="bookTable"></tbody>
          </table>
        </div>
        <p id="bookMessage" class="message" role="status"></p>
      </div>
    </section>

    <!-- LOGIN -->
    <section class="page" id="login">
      <div class="panel">
        <h2>Login to Your Account</h2>
        <p class="muted">Select your role to open the appropriate dashboard.</p>
        <form id="loginForm">
          <div class="form-grid">
            <div class="field">
              <label for="loginRole">Login as</label>
              <select id="loginRole">
                <option value="student">Student</option>
                <option value="admin">Librarian / Admin</option>
              </select>
            </div>
            <div class="field">
              <label for="loginEmail">Email</label>
              <input id="loginEmail" type="email" required placeholder="you@college.edu">
            </div>
            <div class="field full">
              <label for="loginPassword">Password</label>
              <input id="loginPassword" type="password" required minlength="8" placeholder="Enter password">
            </div>
          </div>
          <div class="form-actions"><button class="btn">Login</button></div>
```

```
      </form>
      <p id="loginMessage" class="message" role="status"></p>
      <p class="notice">Demo interface: connect the backend to verify credentials securely. Selecting a role here does not grant real admin permissions.</p>
    </div>
</section>

<!-- REGISTRATION -->
<section class="page" id="register">
    <div class="panel">
      <h2>Student Registration</h2>
      <form id="registerForm">
        <div class="form-grid">
          <div class="field"><label for="regName">Full Name</label><input id="regName" required></div>
          <div class="field"><label for="regEmail">Email</label><input id="regEmail" type="email" required></div>
          <div class="field"><label for="regStudentId">Student ID</label><input id="regStudentId" required></div>
          <div class="field"><label for="regDepartment">Department</label><input id="regDepartment" value="CSE" required></div>
          <div class="field full"><label for="regPassword">Password (at least 8 characters)</label><input id="regPassword" type="password" minlength="8" required></div>
        </div>
        <div class="form-actions"><button class="btn">Register</button></div>
      </form>
      <p id="registerMessage" class="message" role="status"></p>
    </div>
</section>

<!-- STUDENT DASHBOARD -->
<section class="page" id="dashboard">
    <div class="section-head">
      <div><h2>Student Dashboard</h2><p class="muted" id="studentWelcome">Welcome to your library account.</p></div>
    </div>
    <div class="stats">
      <div class="stat"><p>Books Currently Issued</p><strong id="issuedCount">0</strong></div>
      <div class="stat"><p>Overdue Books</p><strong id="overdueCount">0</strong></div>
      <div class="stat"><p>Unpaid Fine (demo)</p><strong id="fineCount">₹0</strong></div>
      <div class="stat"><p>Available Books</p><strong id="availableCount">0</strong></div>
    </div>
    <div class="panel">
      <div class="section-head"><h3>My Borrow History</h3><button class="btn secondary" onclick="renderHistory()">Refresh</button></div>
```

```
      <div class="table-wrap"><table>
        <thead><tr><th>Book</th><th>Issue Date</th><th>Due Date</th><th>Return Date</th><th>Status</th><th>Fine</th></tr></thead>
        <tbody id="historyTable"></tbody>
      </table></div>
    </div>
</section>

<!-- MATERIALS -->
<section class="page" id="materials">
    <div class="section-head"><h2>Study Materials</h2><span class="muted">Academic year and semester</span></div>
    <div class="panel">
      <div class="toolbar">
        <select id="yearFilter" onchange="renderMaterials()">
          <option value="">All Years</option><option>First Year</option><option>Second Year</option><option>Third Year</option><option>Final Year</option>
        </select>
        <select id="semesterFilter" onchange="renderMaterials()">
          <option value="">All Semesters</option>
          <option value="1">Semester 1</option><option value="2">Semester 2</option>
          <option value="3">Semester 3</option><option value="4">Semester 4</option>
          <option value="5">Semester 5</option><option value="6">Semester 6</option>
          <option value="7">Semester 7</option><option value="8">Semester 8</option>
        </select>
      </div>
      <div id="materialGrid" class="material-grid"></div>
      <p class="notice">Only uploaded documents can be downloaded in the connected system. These are sample material entries for the demo.</p>
    </div>
</section>

<!-- PROFILE -->
<section class="page" id="profile">
    <div class="panel">
      <h2>My Profile</h2>
      <form id="profileForm">
        <div class="form-grid">
          <div class="field"><label for="profileName">Full Name</label><input id="profileName" required></div>
          <div class="field"><label for="profileEmail">Email</label><input id="profileEmail" type="email" required></div>
          <div class="field"><label for="profileStudentId">Student ID</label><input id="profileStudentId"></div>
```

```
            <div class="field"><label for="profileDepartment">Department</label><input id="profileDepartment"></div>
          </div>
          <div class="form-actions"><button class="btn">Save Profile (Demo)</button></div>
        </form>
        <p id="profileMessage" class="message" role="status"></p>
    </div>
</section>

<!-- ADMIN -->
<section class="page" id="admin">
    <div class="section-head"><h2>Librarian / Admin Dashboard</h2></div>
    <p class="notice">Demo controls only. In production, every admin operation must be authorized by the Node.js backend.</p>
    <div class="stats">
        <div class="stat"><p>Total Book Titles</p><strong id="adminBookCount">0</strong></div>
        <div class="stat"><p>Total Copies</p><strong id="adminCopies">0</strong></div>
        <div class="stat"><p>Active Issues (demo)</p><strong id="adminIssues">0</strong></div>
        <div class="stat"><p>Registered Demo Users</p><strong id="adminUsers">0</strong></div>
    </div>

    <div class="panel">
        <h3>Add / Update Book</h3>
        <form id="bookForm">
            <input type="hidden" id="editBookId">
            <div class="form-grid">
                <div class="field"><label for="bookTitle">Book Title</label><input id="bookTitle" required></div>
                <div class="field"><label for="bookAuthor">Author</label><input id="bookAuthor" required></div>
                <div class="field"><label for="bookCategory">Category</label><select id="bookCategory"><option>Computer Science</option><option>Programming</option><option>DBMS</option><option>Networking</option><option>General</option></select></div>
                <div class="field"><label for="bookCopies">Total Copies</label><input id="bookCopies" type="number" min="1" value="1" required></div>
            </div>
            <div class="form-actions"><button class="btn">Save Book</button><button type="button" class="btn secondary" onclick="resetBookForm()">Clear</button></div>
        </form>
        <p id="adminBookMessage" class="message" role="status"></p>
    </div>

    <div class="panel">
        <h3>Manage Inventory</h3>
        <div class="table-wrap"><table>
```

```
        <thead><tr><th>Title</th><th>Author</th><th>Total</th><th>Available</th><th>Actions</th></tr></thead>
        <tbody id="adminBookTable"></tbody>
      </table></div>
    </div>

    <div class="panel">
      <h3>Issue Book to Student</h3>
      <form id="issueForm">
        <div class="form-grid">
          <div class="field"><label for="issueStudent">Student Email</label><input id="issueStudent" type="email" required></div>
          <div class="field"><label for="issueBook">Book</label><select id="issueBook" required></select></div>
          <div class="field"><label for="issueDate">Issue Date</label><input id="issueDate" type="date" required></div>
          <div class="field"><label for="dueDate">Expected Return Date</label><input id="dueDate" type="date" required></div>
        </div>
        <div class="form-actions"><button class="btn">Issue Book</button></div>
      </form>
      <p id="issueMessage" class="message" role="status"></p>
    </div>

    <div class="panel">
      <h3>Issued Books, Returns and Fines</h3>
      <p class="muted">Demo overdue fine rate: ₹5 per day. Set your institution's actual policy before production use.</p>
      <div class="table-wrap"><table>
        <thead><tr><th>Student</th><th>Book</th><th>Due Date</th><th>Status</th><th>Fine</th><th>Action</th></tr></thead>
        <tbody id="adminIssueTable"></tbody>
      </table></div>
    </div>

    <div class="panel">
      <h3>Upload Study Material</h3>
      <form id="materialForm">
        <div class="form-grid">
          <div class="field"><label for="materialTitle">Material Title</label><input id="materialTitle" required></div>
          <div class="field"><label for="materialSubject">Subject</label><input id="materialSubject" required></div>
          <div class="field"><label for="materialYear">Academic Year</label><select id="materialYear"><option>First Year</option><option>Second Year</option><option>Third Year</option><option>Final Year</option></select></div>
          <div class="field"><label for="materialSemester">Semester</label><select id="materialSemester"><option value="1">1</option><option value="2">2</option><option value="3">3</option><option value="4">4</option><option value="5">5</option><option value="6">6</option><option value="7">7</option><option value="8">8</option></select></div>
          <div class="field full"><label for="materialFile">Choose PDF or document</label><input id="materialFile" type="file" accept=".pdf,.doc,.docx,.ppt,.pptx" required></div>
```

```
        </div>
        <div class="form-actions"><button class="btn">Add Material (Demo)</button></div>
      </form>
      <p id="materialMessage" class="message" role="status"></p>
    </div>
</section>

<!-- CONTACT -->
<section class="page" id="contact">
    <div class="panel">
      <h2>Contact the Library</h2>
      <p class="muted">Send a message to the librarian.</p>
      <form id="contactForm">
        <div class="form-grid">
          <div class="field"><label for="contactName">Name</label><input id="contactName" required></div>
          <div class="field"><label for="contactEmail">Email</label><input id="contactEmail" type="email" required></div>
          <div class="field full"><label for="contactText">Message</label><textarea id="contactText" rows="5"
required></textarea></div>
        </div>
        <div class="form-actions"><button class="btn">Submit Message</button></div>
      </form>
      <p id="contactMessage" class="message" role="status"></p>
    </div>
</section>
</main>

<footer>
    <p>📚 College Library Management System</p>
    <p>Books • Borrowing • Study Materials • Student Services</p>
</footer>

<script>
// DEMO DATA. This data is held in browser memory and resets on page reload.
let books = [
    {id: 1, title: "Data Structures and Algorithms", author: "Mark Allen Weiss", category: "Computer Science", total: 5, available: 5},
    {id: 2, title: "Database System Concepts", author: "Abraham Silberschatz", category: "DBMS", total: 4, available: 4},
    {id: 3, title: "Operating System Concepts", author: "Abraham Silberschatz", category: "Computer Science", total: 3, available: 3},
    {id: 4, title: "Clean Code", author: "Robert C. Martin", category: "Programming", total: 3, available: 3},
    {id: 5, title: "Computer Networks", author: "Andrew S. Tanenbaum", category: "Networking", total: 2, available: 2},
```

```
    {id: 6, title: "Introduction to Java", author: "Herbert Schildt", category: "Programming", total: 4, available: 4}
];
let materials = [
    {id: 1, title: "Data Structures Notes", subject: "Data Structures", year: "Second Year", semester: 3, file: null},
    {id: 2, title: "DBMS Revision Notes", subject: "Database Systems", year: "Second Year", semester: 4, file: null},
    {id: 3, title: "Operating Systems Notes", subject: "Operating Systems", year: "Third Year", semester: 5, file: null},
    {id: 4, title: "Computer Networks Notes", subject: "Networking", year: "Third Year", semester: 6, file: null},
    {id: 5, title: "Project Guidelines", subject: "Project Work", year: "Final Year", semester: 7, file: null}
];
let users = [];
let issues = [];
let currentUser = null;
let nextBookId = 7;
let nextIssueId = 1;
let nextMaterialId = 6;

function showPage(id) {
    document.querySelectorAll(".page").forEach(p => p.classList.remove("active"));
    const page = document.getElementById(id);
    if (page) page.classList.add("active");
    document.querySelectorAll(".navbar button").forEach(b =>
        b.classList.toggle("active", b.dataset.page === id)
    );
    if (id === "books") renderBooks();
    if (id === "materials") renderMaterials();
    if (id === "dashboard") renderHistory();
    if (id === "admin") renderAdmin();
    if (id === "profile") fillProfile();
    window.scrollTo({top: 0, behavior: "smooth"});
}

document.querySelectorAll(".navbar button").forEach(button => {
    button.addEventListener("click", () => showPage(button.dataset.page));
});

function escapeHTML(value) {
    return String(value).replace(/[&<>"']/g, ch => ({
        "&": "&amp;", "<": "&lt;", ">": "&gt;",
        '"': "&quot;", "'": "&#39;"
```

```
    })[ch]);
}
function money(amount) { return "₹" + Number(amount).toFixed(2); }
function todayString() {
    const d = new Date();
    return `${d.getFullYear()}-${String(d.getMonth()+1).padStart(2,"0")}-${String(d.getDate()).padStart(2,"0")}`;
}
function dateDiff(start, end) {
    return Math.max(0, Math.floor((new Date(end + "T00:00:00") - new Date(start + "T00:00:00")) / 86400000));
}
function fineFor(issue) {
    const end = issue.returnDate || todayString();
    return Math.max(0, dateDiff(issue.dueDate, end)) * 5;
}

function renderFeatured() {
    document.getElementById("featuredBooks").innerHTML = books.slice(0,3).map(book => `
        <article class="book-card">
            <div class="book-cover">📖<br>${escapeHTML(book.category)}</div>
            <div class="book-info">
                <h3>${escapeHTML(book.title)}</h3>
                <p>${escapeHTML(book.author)}</p>
                <button class="btn" onclick="showDetails(${book.id})">View Details</button>
            </div>
        </article>`).join("");
}
function showDetails(id) {
    const b = books.find(book => book.id === id);
    if (!b) return;
    document.getElementById("bookMessage").textContent =
        `${b.title} | Author: ${b.author} | Category: ${b.category} | Available copies: ${b.available} of ${b.total}.`;
    showPage("books");
}
function searchFromHome() {
    document.getElementById("catalogSearch").value =
        document.getElementById("homeSearch").value;
    showPage("books");
    renderBooks();
}
```

```
function renderBooks() {
  const q = (document.getElementById("catalogSearch").value || "").toLowerCase().trim();
  const category = document.getElementById("categoryFilter").value;
  const list = books.filter(b =>
    (b.title.toLowerCase().includes(q) ||
     b.author.toLowerCase().includes(q) ||
     b.category.toLowerCase().includes(q)) &&
    (!category || b.category === category)
  );
  document.getElementById("bookTable").innerHTML = list.length ? list.map(b => `
    <tr>
      <td>${escapeHTML(b.title)}</td>
      <td>${escapeHTML(b.author)}</td>
      <td>${escapeHTML(b.category)}</td>
      <td><span class="badge ${b.available === 0 ? "unavailable" : ""}">${b.available > 0 ? b.available + " available" : "Unavailable"}</span></td>
      <td><button class="btn small" onclick="showDetails(${b.id})">Details</button></td>
    </tr>`).join("") : '<tr><td colspan="5" class="empty">No matching books found.</td></tr>';
}
function renderAdmin() {
  document.getElementById("adminBookCount").textContent = books.length;
  document.getElementById("adminCopies").textContent = books.reduce((n,b)=>n+b.total,0);
  document.getElementById("adminIssues").textContent = issues.filter(i=>!i.returnDate).length;
  document.getElementById("adminUsers").textContent = users.length;
  document.getElementById("adminBookTable").innerHTML = books.map(b => `
    <tr>
      <td>${escapeHTML(b.title)}</td><td>${escapeHTML(b.author)}</td>
      <td>${b.total}</td><td>${b.available}</td>
      <td>
        <button class="btn small secondary" onclick="editBook(${b.id})">Edit</button>
        <button class="btn small danger" onclick="deleteBook(${b.id})">Delete</button>
      </td>
    </tr>`).join("");
  document.getElementById("issueBook").innerHTML =
    books.map(b => `<option value="${b.id}" ${b.available < 1 ? "disabled" : ""}>${escapeHTML(b.title)} (${b.available} available)</option>`).join("");
  document.getElementById("adminIssueTable").innerHTML = issues.length ? issues.map(i => `
    <tr>
      <td>${escapeHTML(i.email)}</td><td>${escapeHTML(books.find(b=>b.id===i.bookId)?.title || "Deleted book")}</td>
      <td>${escapeHTML(i.dueDate)}</td><td>${i.returnDate ? "Returned" : "Issued"}</td>
```

```
            <td>${money(fineFor(i))}</td>
            <td>${i.returnDate ? "—" : `<button class="btn small success" onclick="returnBook(${i.id})">Return</button>`}</td>
        </tr>`).join("") : '<tr><td colspan="6" class="empty">No issued books yet.</td></tr>';
}
function resetBookForm() {
    document.getElementById("bookForm").reset();
    document.getElementById("editBookId").value = "";
}
document.getElementById("bookForm").addEventListener("submit", event => {
    event.preventDefault();
    const id = Number(document.getElementById("editBookId").value);
    const title = document.getElementById("bookTitle").value.trim();
    const author = document.getElementById("bookAuthor").value.trim();
    const category = document.getElementById("bookCategory").value;
    const total = Number(document.getElementById("bookCopies").value);
    if (!title || !author || !Number.isInteger(total) || total < 1) return;
    if (id) {
        const b = books.find(x => x.id === id);
        const alreadyIssued = b.total - b.available;
        if (total < alreadyIssued) {
            document.getElementById("adminBookMessage").textContent =
                "Total copies cannot be less than copies already issued.";
            return;
        }
        Object.assign(b, {title, author, category, total, available: total - alreadyIssued});
        document.getElementById("adminBookMessage").textContent = "Book updated in demo.";
    } else {
        books.push({id: nextBookId++, title, author, category, total, available: total});
        document.getElementById("adminBookMessage").textContent = "Book added to demo catalog.";
    }
    resetBookForm(); renderAll();
});
function editBook(id) {
    const b = books.find(x => x.id === id);
    if (!b) return;
    document.getElementById("editBookId").value = b.id;
    document.getElementById("bookTitle").value = b.title;
    document.getElementById("bookAuthor").value = b.author;
    document.getElementById("bookCategory").value = b.category;
```

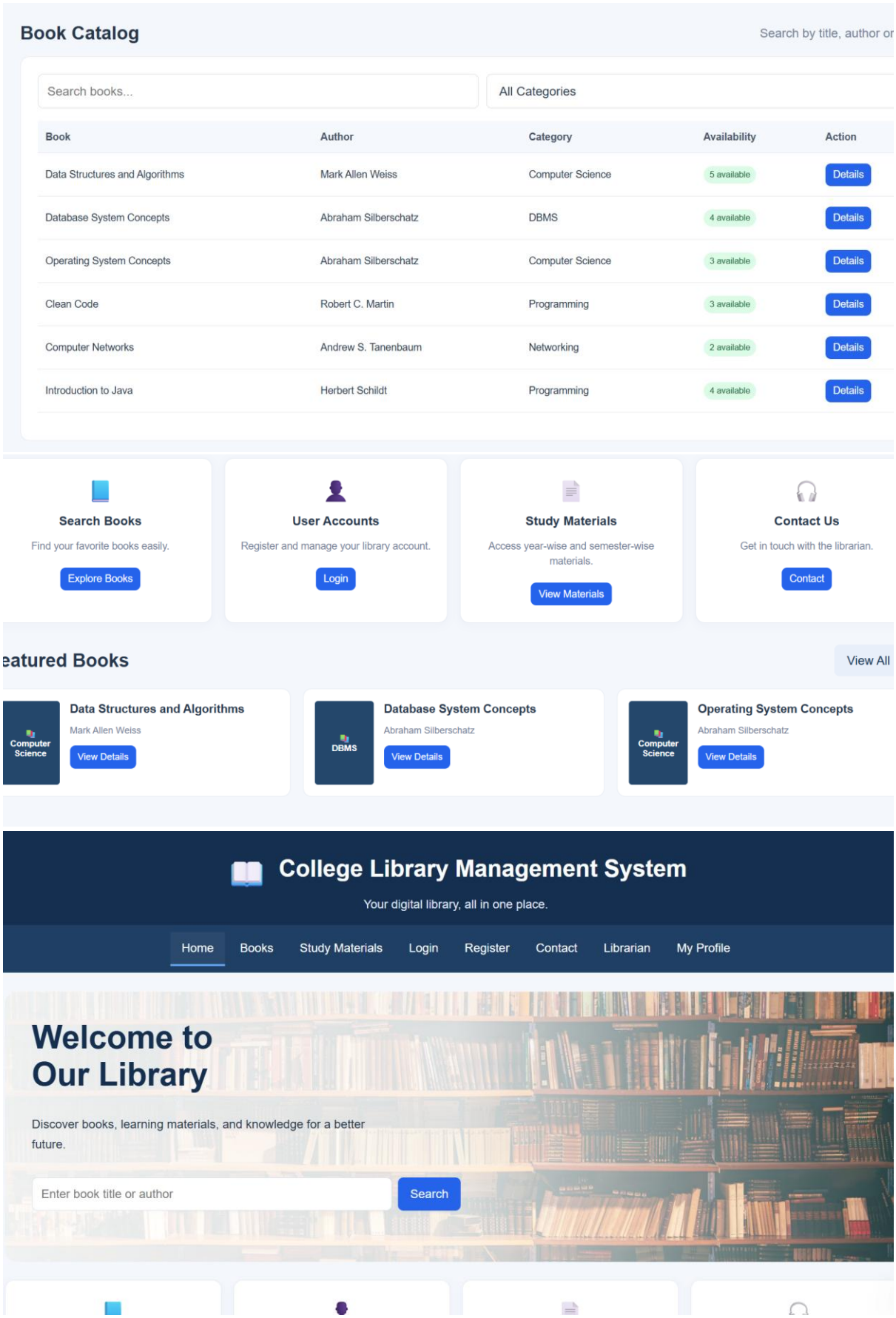
```
    document.getElementById("bookCopies").value = b.total;
    window.scrollTo({top: document.getElementById("bookForm").offsetTop, behavior: "smooth"});
}
function deleteBook(id) {
    const b = books.find(x => x.id === id);
    if (!b) return;
    if (b.available !== b.total) {
        alert("Return all issued copies before deleting this book.");
        return;
    }
    if (confirm("Delete " + b.title + " from the demo catalog?")) {
        books = books.filter(x => x.id !== id);
        renderAll();
    }
}
document.getElementById("issueDate").value = todayString();
document.getElementById("dueDate").value = (() => {
    const d = new Date(); d.setDate(d.getDate()+14);
    return `${d.getFullYear()}-${String(d.getMonth()+1).padStart(2,"0")}-${String(d.getDate()).padStart(2,"0")}`;
})();
document.getElementById("issueForm").addEventListener("submit", event => {
    event.preventDefault();
    const email = document.getElementById("issueStudent").value.trim().toLowerCase();
    const bookId = Number(document.getElementById("issueBook").value);
    const issueDate = document.getElementById("issueDate").value;
    const dueDate = document.getElementById("dueDate").value;
    const b = books.find(x=>x.id===bookId);
    if (!b || b.available < 1) {
        document.getElementById("issueMessage").textContent = "No copies available.";
        return;
    }
    if (!email || !issueDate || !dueDate || dueDate < issueDate) {
        document.getElementById("issueMessage").textContent = "Check the student email and dates.";
        return;
    }
    b.available--;
    issues.push({id: nextIssueId++, email, bookId, issueDate, dueDate, returnDate: null});
    document.getElementById("issueMessage").textContent =
        "Book issued in demo. Fine is calculated at ₹5 per overdue day.";
```

```
    renderAll();
});
function returnBook(id) {
    const issue = issues.find(i=>i.id===id);
    if (!issue || issue.returnDate) return;
    issue.returnDate = todayString();
    const b = books.find(x=>x.id===issue.bookId);
    if (b) b.available = Math.min(b.total, b.available+1);
    alert("Book marked as returned. Demo fine: " + money(fineFor(issue)));
    renderAll();
}
function renderHistory() {
    const email = currentUser?.email?.toLowerCase();
    const list = issues.filter(i => email && i.email === email);
    document.getElementById("issuedCount").textContent = list.filter(i=>!i.returnDate).length;
    document.getElementById("overdueCount").textContent = list.filter(i=>!i.returnDate && i.dueDate < todayString()).length;
    document.getElementById("fineCount").textContent = money(list.reduce((n,i)=>n+fineFor(i),0));
    document.getElementById("availableCount").textContent = books.reduce((n,b)=>n+b.available,0);
    document.getElementById("historyTable").innerHTML = list.length ? list.map(i => `
        <tr><td>${escapeHTML(books.find(b=>b.id===i.bookId)?.title || "Deleted book")}</td>
        <td>${escapeHTML(i.issueDate)}</td><td>${escapeHTML(i.dueDate)}</td>
        <td>${escapeHTML(i.returnDate || "—")}</td><td>${i.returnDate ? "Returned" : "Issued"}</td>
        <td>${money(fineFor(i))}</td></tr>`).join("") :
        '<tr><td colspan="6" class="empty">No borrowing records for this demo account.</td></tr>';
}
document.getElementById("loginForm").addEventListener("submit", event => {
    event.preventDefault();
    const email = document.getElementById("loginEmail").value.trim().toLowerCase();
    const role = document.getElementById("loginRole").value;
    // UI demonstration only: NOT secure authentication.
    currentUser = users.find(u=>u.email.toLowerCase()===email) || {
        name: email.split("@")[0], email, role
    };
    currentUser.role = role;
    document.getElementById("loginMessage").textContent =
        "Demo login only. Connect the backend for password verification.";
    if (role === "admin") showPage("admin");
    else {
        document.getElementById("studentWelcome").textContent = "Welcome, " + currentUser.name;
```

```
        showPage("dashboard");
    }
    renderHistory();
});
document.getElementById("registerForm").addEventListener("submit", event => {
    event.preventDefault();
    const name = document.getElementById("regName").value.trim();
    const email = document.getElementById("regEmail").value.trim().toLowerCase();
    const studentId = document.getElementById("regStudentId").value.trim();
    const department = document.getElementById("regDepartment").value.trim();
    const password = document.getElementById("regPassword").value;
    if (users.some(u=>u.email.toLowerCase()===email)) {
        document.getElementById("registerMessage").textContent = "This email is already registered in the demo.";
        return;
    }
    users.push({name,email,studentId,department});
    currentUser = {name,email,studentId,department};
    document.getElementById("registerMessage").textContent =
        "Demo registration complete. No account was saved to a database.";
    fillProfile();
});
function fillProfile() {
    if (!currentUser) return;
    document.getElementById("profileName").value = currentUser.name || "";
    document.getElementById("profileEmail").value = currentUser.email || "";
    document.getElementById("profileStudentId").value = currentUser.studentId || "";
    document.getElementById("profileDepartment").value = currentUser.department || "";
}
document.getElementById("profileForm").addEventListener("submit", event => {
    event.preventDefault();
    if (!currentUser) {
        document.getElementById("profileMessage").textContent = "Register or log in first in this demo.";
        return;
    }
    currentUser.name = document.getElementById("profileName").value.trim();
    currentUser.email = document.getElementById("profileEmail").value.trim();
    currentUser.studentId = document.getElementById("profileStudentId").value.trim();
    currentUser.department = document.getElementById("profileDepartment").value.trim();
    document.getElementById("profileMessage").textContent = "Profile updated in demo memory.";
```

```
});
function renderMaterials() {
  const year = document.getElementById("yearFilter").value;
  const sem = document.getElementById("semesterFilter").value;
  const list = materials.filter(m=>(!year || m.year===year) && (!sem || String(m.semester)===sem));
  document.getElementById("materialGrid").innerHTML = list.length ? list.map(m=>`
    <article class="material">
      <div style="font-size:28px">📄</div>
      <h3>${escapeHTML(m.title)}</h3>
      <p>${escapeHTML(m.subject)}</p>
      <p class="muted">${escapeHTML(m.year)} • Semester ${m.semester}</p>
      ${m.file ? `<button class="btn small" onclick="downloadMaterial(${m.id})">Download</button>` : '<span class="badge unavailable">Sample only</span>'}
    </article>`).join("") : '<p class="empty">No materials for this selection.</p>';
}
function downloadMaterial(id) {
  const m = materials.find(x=>x.id===id);
  if (!m || !m.file) return;
  const url = URL.createObjectURL(m.file);
  const a = document.createElement("a");
  a.href = url; a.download = m.file.name; a.click();
  URL.revokeObjectURL(url);
}
document.getElementById("materialForm").addEventListener("submit", event => {
  event.preventDefault();
  const file = document.getElementById("materialFile").files[0];
  if (!file) return;
  const allowed = /\.(pdf|doc|docx|ppt|pptx)$/i.test(file.name);
  if (!allowed) {
    document.getElementById("materialMessage").textContent = "Choose a PDF, DOC, DOCX, PPT or PPTX file.";
    return;
  }
  materials.push({
    id: nextMaterialId++,
    title: document.getElementById("materialTitle").value.trim(),
    subject: document.getElementById("materialSubject").value.trim(),
    year: document.getElementById("materialYear").value,
    semester: Number(document.getElementById("materialSemester").value),
```

SAMPLE OUTPUT:



EXP 2:

AIM:

To design and develop an Online Hotel Booking System using HTML, CSS, and JavaScript that allows users to view hotels, select room types, check availability, calculate booking costs, and confirm reservations.

CODE:

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <title>Hotel Booking System</title>
  <style>
    body {
      font-family: Arial, sans-serif;
      margin: 0;
      background: #f2f5f9;
      color: #243247;
    }
    header {
      background: #173b5e;
      color: white;
      padding: 25px;
      text-align: center;
    }
    nav {
      background: #244e73;
      padding: 14px;
      text-align: center;
```

```
}
nav a {
  color: white;
  text-decoration: none;
  margin: 12px;
}
main {
  max-width: 900px;
  margin: 25px auto;
  padding: 15px;
}
section {
  background: white;
  padding: 22px;
  margin-bottom: 20px;
  border-radius: 10px;
  box-shadow: 0 2px 8px #0001;
}
input, select, textarea {
  width: 100%;
  padding: 11px;
  margin: 8px 0 14px;
  box-sizing: border-box;
}
button {
  background: #2563eb;
  color: white;
  border: 0;
  padding: 11px 17px;
  border-radius: 5px;
  cursor: pointer;
}
```

```
        .rooms {
            display: grid;
            grid-template-columns: repeat(auto-fit, minmax(200px, 1fr));
            gap: 15px;
        }
        .room {
            border: 1px solid #ddd;
            border-radius: 8px;
            padding: 15px;
        }
        footer {
            background: #173b5e;
            color: white;
            text-align: center;
            padding: 18px;
        }
    </style>
</head>
<body>

<header>
    <h1>🏨 Grand Stay Hotel</h1>
    <p>Comfort, luxury and memorable stays</p>
</header>

<nav>
    <a href="#search">Search</a>
    <a href="#rooms">Rooms</a>
    <a href="#booking">Booking</a>
    <a href="#login">Login</a>
    <a href="#contact">Contact</a>
</nav>
```

```
<main>

<section id="search">
    <h2>Find Your Stay</h2>
    <label>Destination</label>
    <input id="destination" placeholder="Enter destination">

    <label>Number of Guests</label>
    <input id="guests" type="number" min="1" placeholder="Number of guests">

    <button onclick="searchHotel()">Search Hotels</button>
</section>

<section id="rooms">
    <h2>Choose Your Room</h2>
    <div class="rooms">
        <div class="room">
            <h3>Standard Room</h3>
            <p>Comfortable room for your stay.</p>
            <p>₹2,000 / night</p>
            <button onclick="selectRoom('Standard Room')">Select Room</button>
        </div>
        <div class="room">
            <h3>Deluxe Room</h3>
            <p>Spacious room with premium comfort.</p>
            <p>₹3,500 / night</p>
            <button onclick="selectRoom('Deluxe Room')">Select Room</button>
        </div>
        <div class="room">
            <h3>Suite Room</h3>
            <p>Luxury accommodation and extra space.</p>
```

```
            <p>₹5,000 / night</p>
            <button onclick="selectRoom('Suite Room')">Select Room</button>
        </div>
    </div>
</section>

<section id="booking">
    <h2>Book Your Room</h2>
    <form onsubmit="makeBooking(event)">
        <label>Room Type</label>
        <select id="roomType" required>
            <option value="">Select a room</option>
            <option>Standard Room</option>
            <option>Deluxe Room</option>
            <option>Suite Room</option>
        </select>

        <label>Check-in Date</label>
        <input id="bookingCheckin" type="date" required>

        <label>Check-out Date</label>
        <input id="bookingCheckout" type="date" required>

        <button type="submit">Book Now</button>
    </form>
    <button onclick="changeBooking()">Modify Booking</button>
    <button onclick="cancelBooking()">Cancel Booking</button>
</section>

<section id="login">
    <h2>Login</h2>
    <form onsubmit="login(event)">
```

```
        <input type="email" placeholder="Email address" required>
        <input type="password" placeholder="Password" required>
        <button type="submit">Login</button>
    </form>
</section>

<section id="register">
    <h2>Create an Account</h2>
    <form onsubmit="registerUser(event)">
        <input type="text" placeholder="Full name" required>
        <input type="email" placeholder="Email address" required>
        <input type="password" placeholder="Create password" required>
        <button type="submit">Register</button>
    </form>
</section>

<section id="payment">
    <h2>Payment</h2>
    <p>Continue to confirm your booking.</p>
    <form onsubmit="payment(event)">
        <label>Payment Method</label>
        <select required>
            <option value="">Choose payment method</option>
            <option>UPI</option>
            <option>Credit / Debit Card</option>
            <option>Pay at Hotel</option>
        </select>
        <button type="submit">Confirm Payment (Demo)</button>
    </form>
</section>

<section id="profile">
```

```
    <h2>My Profile</h2>
    <button onclick="editProfile()">Edit Profile</button>
</section>

<section id="contact">
    <h2>Contact Us</h2>
    <form onsubmit="contactForm(event)">
        <input type="text" placeholder="Your name" required>
        <input type="email" placeholder="Your email" required>
        <textarea placeholder="Your message" required></textarea>
        <button type="submit">Send Message</button>
    </form>
</section>

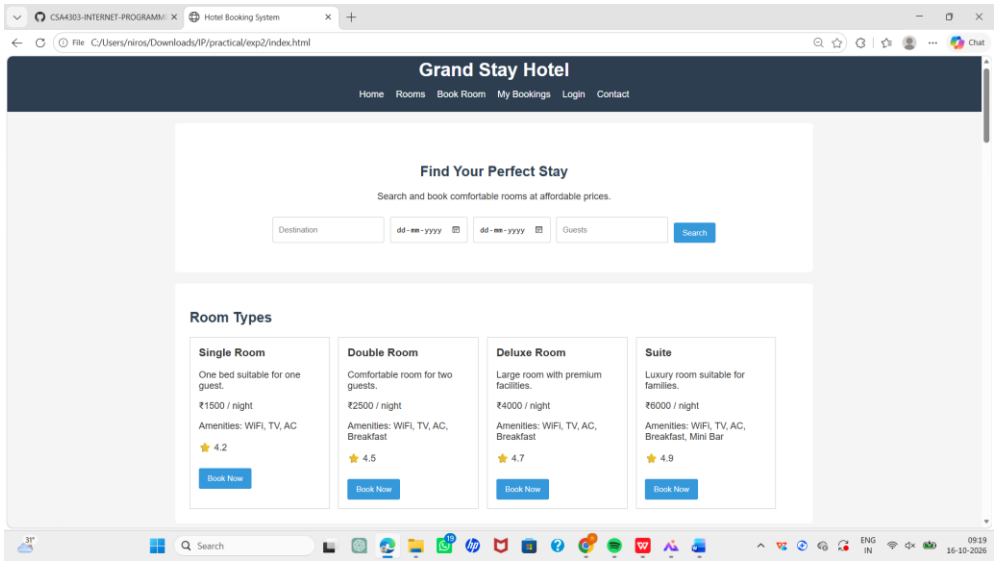
<section id="admin">
    <h2>Admin Dashboard</h2>
    <button onclick="adminAction('Manage Rooms')">Manage Rooms</button>
    <button onclick="adminAction('View Bookings')">View Bookings</button>
    <button onclick="adminAction('Manage Customers')">Manage Customers</button>
</section>

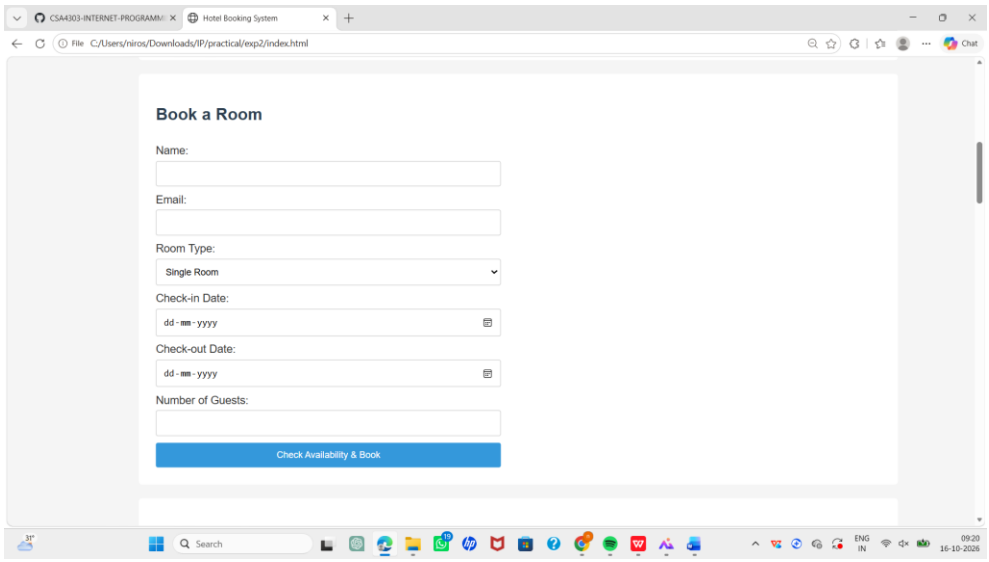
</main>

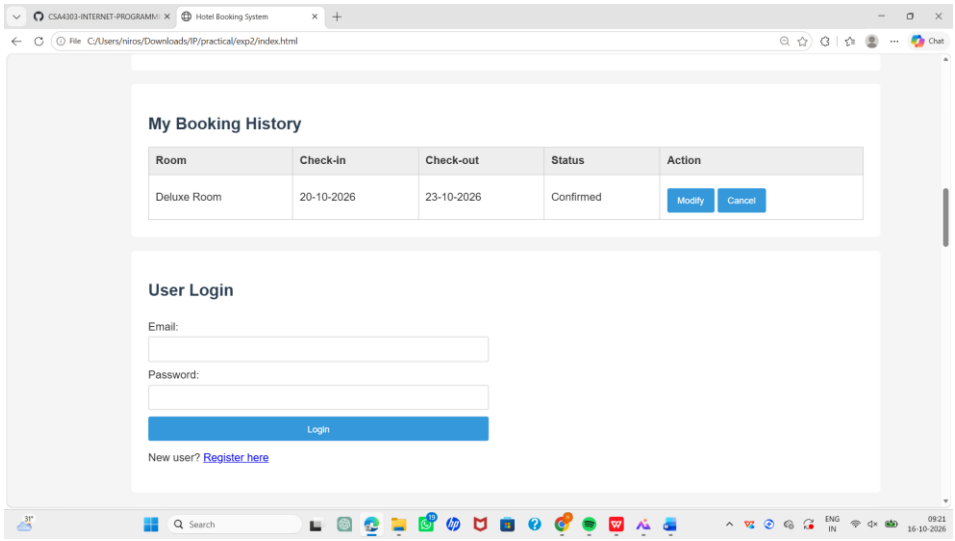
<footer>
    <p>© 2026 Grand Stay Hotel | Hotel Booking System</p>
</footer>

<script src="script.js"></script>
</body>
</html>
```

SAMPLE OUTPUT:







EXP 3:

AIM:

To design and develop an online Bus Ticket Booking System using HTML, CSS, and JavaScript that allows users to search buses, select seats, enter passenger details, calculate ticket fares, and confirm their bookings.

CODE:

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <title>Hotel Booking System</title>
  <style>
    body {
      font-family: Arial, sans-serif;
      margin: 0;
      background: #f2f5f9;
      color: #243247;
    }
    header {
      background: #173b5e;
      color: white;
      text-align: center;
      padding: 25px;
    }
    nav {
      background: #244e73;
      padding: 15px;
      text-align: center;
    }
```

```
nav a {
    color: white;
    margin: 12px;
    text-decoration: none;
}
main {
    max-width: 850px;
    margin: 25px auto;
    padding: 15px;
}
section {
    background: white;
    padding: 22px;
    margin-bottom: 20px;
    border-radius: 10px;
}
input, select, textarea {
    width: 100%;
    padding: 12px;
    margin: 8px 0 15px;
    box-sizing: border-box;
}
button {
    background: #2563eb;
    color: white;
    padding: 11px 16px;
    border: none;
    border-radius: 5px;
    cursor: pointer;
}
.rooms {
    display: grid;
```

```
            grid-template-columns: repeat(auto-fit, minmax(200px, 1fr));
            gap: 15px;
        }
        .room {
            border: 1px solid #ddd;
            padding: 15px;
            border-radius: 8px;
        }
        footer {
            text-align: center;
            background: #173b5e;
            color: white;
            padding: 18px;
        }
    </style>
</head>
<body>

<header>
    <h1>Grand Stay Hotel</h1>
    <p>Comfort, Luxury and Memorable Stays</p>
</header>

<nav>
    <a href="#search">Search</a>
    <a href="#rooms">Rooms</a>
    <a href="#booking">Booking</a>
    <a href="#login">Login</a>
    <a href="#contact">Contact</a>
</nav>

<main>
```

```
<section id="search">
    <h2>Search Hotels</h2>
    <input id="destination" placeholder="Enter destination" required>
    <input id="guests" type="number" min="1"
          placeholder="Number of guests" required>
    <button onclick="searchHotel()">Search Hotels</button>
</section>

<section id="rooms">
    <h2>Available Room Types</h2>
    <div class="rooms">
        <div class="room">
            <h3>Standard Room</h3>
            <p>₹2,000 per night</p>
            <button onclick="selectRoom('Standard Room')">
                Select Room
            </button>
        </div>
        <div class="room">
            <h3>Deluxe Room</h3>
            <p>₹3,500 per night</p>
            <button onclick="selectRoom('Deluxe Room')">
                Select Room
            </button>
        </div>
        <div class="room">
            <h3>Suite Room</h3>
            <p>₹5,000 per night</p>
            <button onclick="selectRoom('Suite Room')">
                Select Room
            </button>
        </div>
```

```
    </div>
</section>

<section id="booking">
    <h2>Book Your Room</h2>
    <form onsubmit="makeBooking(event)">
        <label for="roomType">Room Type</label>
        <select id="roomType" required>
            <option value="">Select a room</option>
            <option>Standard Room</option>
            <option>Deluxe Room</option>
            <option>Suite Room</option>
        </select>

        <label for="bookingCheckin">Check-in</label>
        <input type="date" id="bookingCheckin" required>

        <label for="bookingCheckout">Check-out</label>
        <input type="date" id="bookingCheckout" required>

        <button type="submit">Book Now</button>
    </form>
    <button onclick="changeBooking()">Modify Booking</button>
    <button onclick="cancelBooking()">Cancel Booking</button>
</section>

<section id="login">
    <h2>Login</h2>
    <form onsubmit="login(event)">
        <input type="email" placeholder="Email address" required>
        <input type="password" placeholder="Password" required>
        <button type="submit">Login</button>
```

```
    </form>
</section>

<section id="register">
    <h2>Register</h2>
    <form onsubmit="registerUser(event)">
        <input type="text" placeholder="Full name" required>
        <input type="email" placeholder="Email address" required>
        <input type="password" placeholder="Create password" required>
        <button type="submit">Register</button>
    </form>
</section>

<section id="payment">
    <h2>Payment</h2>
    <form onsubmit="payment(event)">
        <select required>
            <option value="">Select payment method</option>
            <option>UPI</option>
            <option>Credit / Debit Card</option>
            <option>Pay at Hotel</option>
        </select>
        <button type="submit">Confirm Payment</button>
    </form>
</section>

<section id="profile">
    <h2>My Profile</h2>
    <button onclick="editProfile()">Edit Profile</button>
</section>

<section id="contact">
```

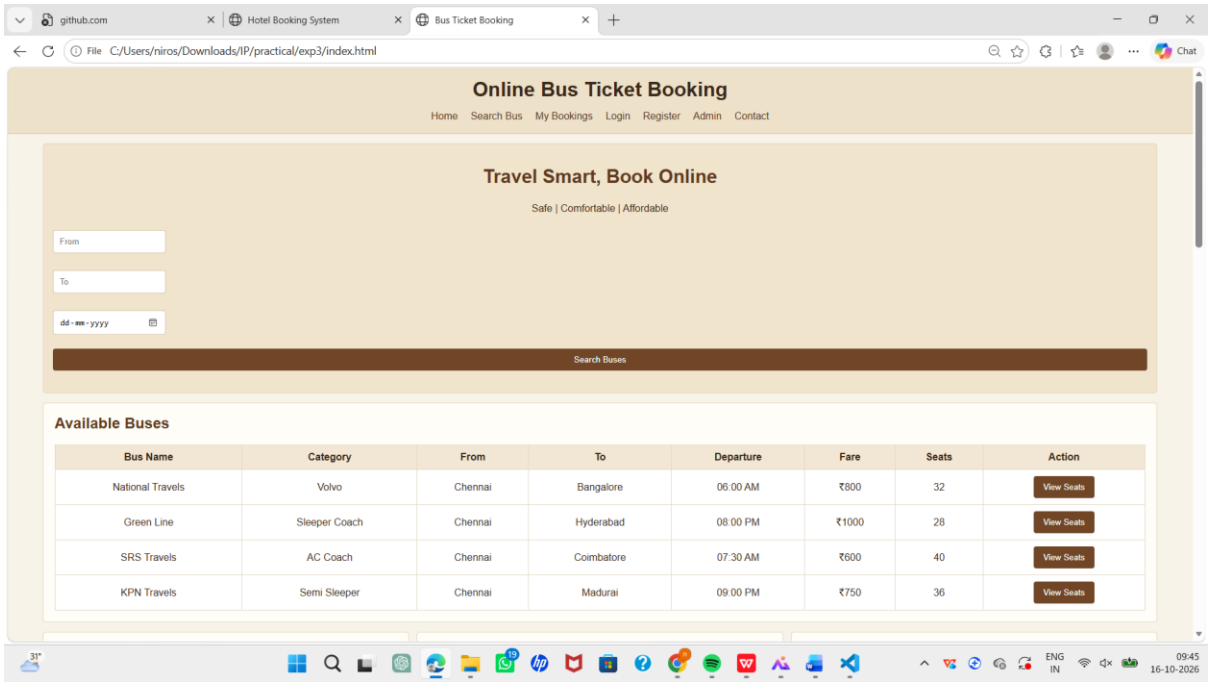
```
        <h2>Contact Us</h2>
        <form onsubmit="contactForm(event)">
            <input type="text" placeholder="Your name" required>
            <input type="email" placeholder="Your email" required>
            <textarea placeholder="Your message" required></textarea>
            <button type="submit">Send Message</button>
        </form>
    </section>

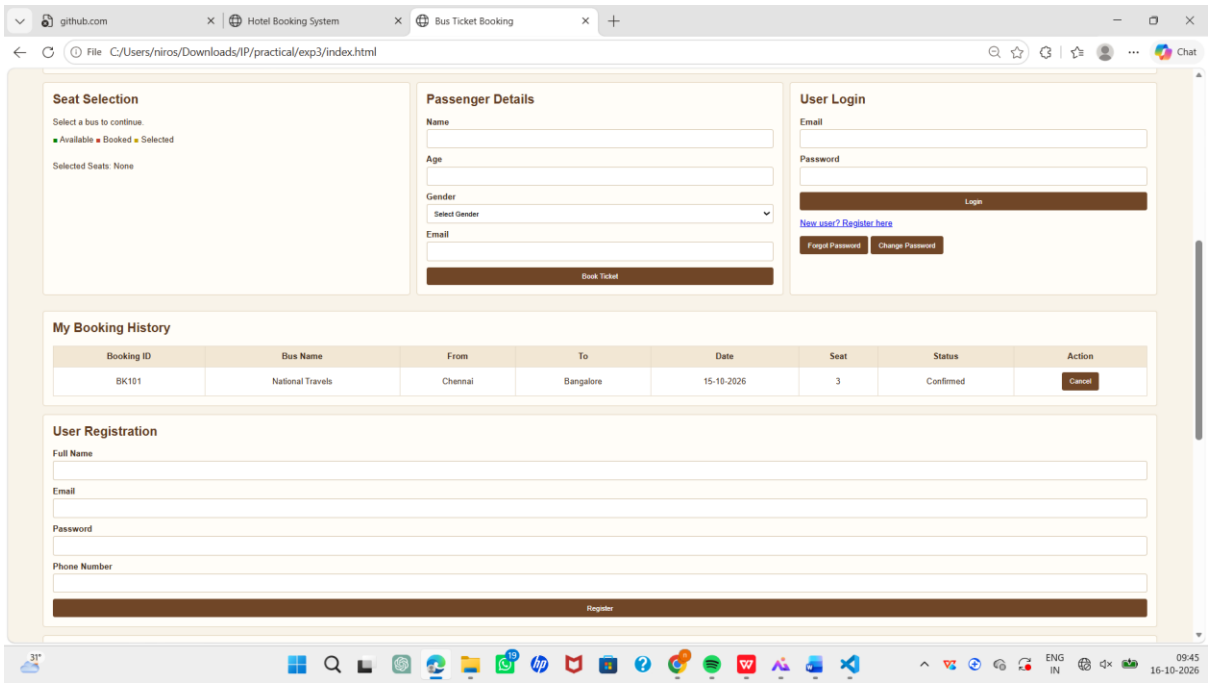
    <section id="admin">
        <h2>Admin Dashboard</h2>
        <button onclick="adminAction('Manage Rooms')">Manage Rooms</button>
        <button onclick="adminAction('View Bookings')">View Bookings</button>
        <button onclick="adminAction('Manage Customers')">Manage Customers</button>
    </section>
</main>

<footer>
    <p>Grand Stay Hotel | Hotel Booking System</p>
</footer>

<script src="script.js"></script>
</body>
</html>
```

SAMPLE OUTPUT:





EXP 4:

AIM:

To design and develop an Online Flight Reservation System using HTML, CSS, and JavaScript that allows users to select departure and destination cities, choose a travel date and class, enter passenger details, calculate ticket fare, and display booking confirmation.

CODE:

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <title>Online Flight Reservation System</title>

  <style>
    body {
      font-family: Arial, sans-serif;
      margin: 0;
      background: #f0f4f8;
      color: #243247;
    }

    header {
      background: #12345a;
      color: white;
      text-align: center;
      padding: 25px;
    }

    main {
```

```
    max-width: 750px;
    margin: 30px auto;
    padding: 20px;
}

section {
    background: white;
    padding: 25px;
    margin-bottom: 20px;
    border-radius: 12px;
    box-shadow: 0 3px 10px #0001;
}

h2 {
    color: #12345a;
}

label {
    display: block;
    margin-top: 14px;
    font-weight: bold;
}

input, select {
    width: 100%;
    padding: 12px;
    margin-top: 7px;
    border: 1px solid #ccd5e0;
    border-radius: 6px;
    box-sizing: border-box;
}
```

```
        button {
            margin-top: 20px;
            padding: 13px 20px;
            background: #2563eb;
            color: white;
            border: none;
            border-radius: 6px;
            cursor: pointer;
        }

        button:hover {
            background: #1d4ed8;
        }

        #confirmation {
            display: none;
            border-left: 5px solid #16a34a;
        }

        footer {
            text-align: center;
            background: #12345a;
            color: white;
            padding: 15px;
        }
    </style>
</head>

<body>

<header>
    <h1>✈ Online Flight Reservation System</h1>
```

```
    <p>Book your journey with ease</p>
</header>

<main>

    <section>
        <h2>Flight Search</h2>

        <form id="flightForm">

            <label for="departure">Departure City</label>
            <select id="departure" required>
                <option value="">Select departure city</option>
                <option>Chennai</option>
                <option>Bengaluru</option>
                <option>Hyderabad</option>
                <option>Mumbai</option>
                <option>Delhi</option>
            </select>

            <label for="destination">Destination City</label>
            <select id="destination" required>
                <option value="">Select destination city</option>
                <option>Chennai</option>
                <option>Bengaluru</option>
                <option>Hyderabad</option>
                <option>Mumbai</option>
                <option>Delhi</option>
            </select>

            <label for="travelDate">Travel Date</label>
            <input type="date" id="travelDate" required>
```

```
        <label for="travelClass">Travel Class</label>
        <select id="travelClass" required>
            <option value="Economy">Economy</option>
            <option value="Business">Business</option>
            <option value="First Class">First Class</option>
        </select>

        <label for="passengers">Number of Passengers</label>
        <input type="number" id="passengers"
               min="1" max="9" value="1" required>

        <button type="submit">Calculate Fare</button>
    </form>

    <p id="fareResult"></p>
</section>

<section>
    <h2>Passenger Details</h2>

    <form id="passengerForm">
        <label for="fullName">Full Name</label>
        <input type="text" id="fullName" required>

        <label for="email">Email Address</label>
        <input type="email" id="email" required>

        <label for="phone">Phone Number</label>
        <input type="tel" id="phone"
               pattern="[0-9]{10}"
               placeholder="10-digit mobile number"
```

```
                required>

            <button type="submit">Confirm Booking</button>
        </form>
    </section>

    <section id="confirmation">
        <h2>Booking Confirmation</h2>
        <div id="bookingDetails"></div>
    </section>

</main>

<footer>
    <p>© 2026 Online Flight Reservation System</p>
</footer>

<script>
    let flight = null;

    const dateInput = document.getElementById("travelDate");
    const today = new Date();
    dateInput.min = [
        today.getFullYear(),
        String(today.getMonth() + 1).padStart(2, "0"),
        String(today.getDate()).padStart(2, "0")
    ].join("-");

    document.getElementById("flightForm").addEventListener("submit", function(event) {
        event.preventDefault();

        const departure = document.getElementById("departure").value;
```

```
const destination = document.getElementById("destination").value;
const date = dateInput.value;
const travelClass = document.getElementById("travelClass").value;
const passengers = Number(document.getElementById("passengers").value);

if (departure === destination) {
    alert("Departure and destination cities must be different.");
    return;
}

if (!date || date < dateInput.min) {
    alert("Please select a valid travel date.");
    return;
}

const baseFare = 3000;
const classMultiplier = {
    "Economy": 1,
    "Business": 2.5,
    "First Class": 4
};

const totalFare = baseFare * classMultiplier[travelClass] * passengers;

flight = {
    departure,
    destination,
    date,
    travelClass,
    passengers,
    totalFare
};
```

```
    document.getElementById("fareResult").textContent =
        "Estimated total fare: ₹" + totalFare.toLocaleString("en-IN");

    document.getElementById("passengerForm").scrollIntoView({
        behavior: "smooth"
    });
});

document.getElementById("passengerForm").addEventListener("submit", function(event) {
    event.preventDefault();

    if (!flight) {
        alert("Please search for a flight and calculate the fare first.");
        return;
    }

    const name = document.getElementById("fullName").value.trim();
    const email = document.getElementById("email").value.trim();
    const phone = document.getElementById("phone").value.trim();

    const bookingId = "FL" + Date.now().toString().slice(-8);

    document.getElementById("bookingDetails").innerHTML = `
        <p><strong>Booking ID:</strong> ${bookingId}</p>
        <p><strong>Passenger:</strong> ${escapeHTML(name)}</p>
        <p><strong>Email:</strong> ${escapeHTML(email)}</p>
        <p><strong>Phone:</strong> ${escapeHTML(phone)}</p>
        <p><strong>From:</strong> ${escapeHTML(flight.departure)}</p>
        <p><strong>To:</strong> ${escapeHTML(flight.destination)}</p>
        <p><strong>Travel Date:</strong> ${escapeHTML(flight.date)}</p>
        <p><strong>Class:</strong> ${escapeHTML(flight.travelClass)}</p>
```

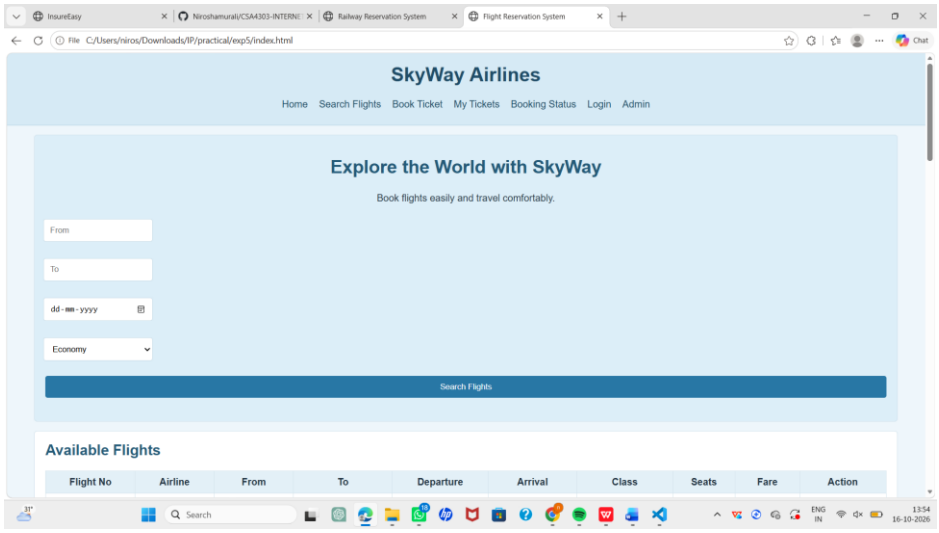
```
            <p><strong>Passengers:</strong> ${flight.passengers}</p>
            <h3>Total Fare: ₹${flight.totalFare.toLocaleString("en-IN")}</h3>
            <p>Your demo booking has been confirmed!</p>
            <button onclick="window.print()">Print Confirmation</button>
        `;

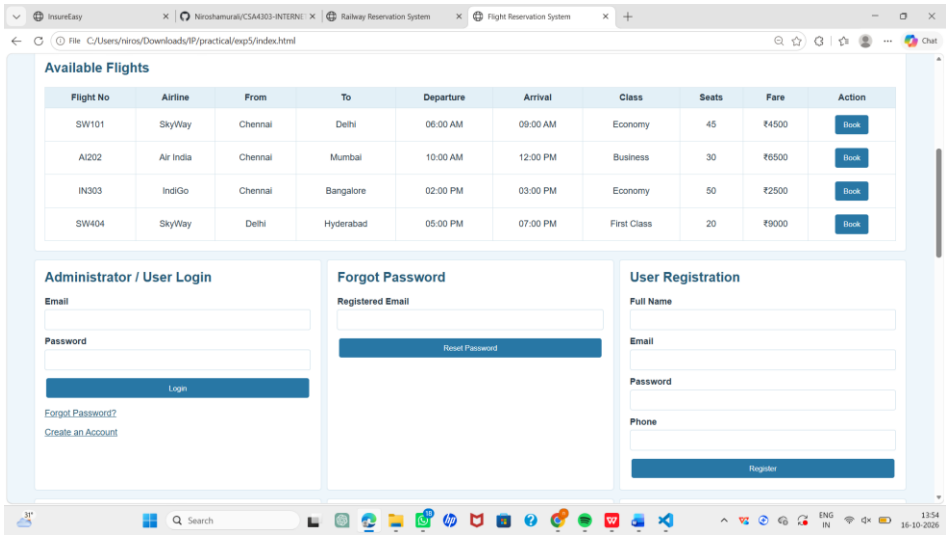
        document.getElementById("confirmation").style.display = "block";
        document.getElementById("confirmation").scrollIntoView({
            behavior: "smooth"
        });
    });

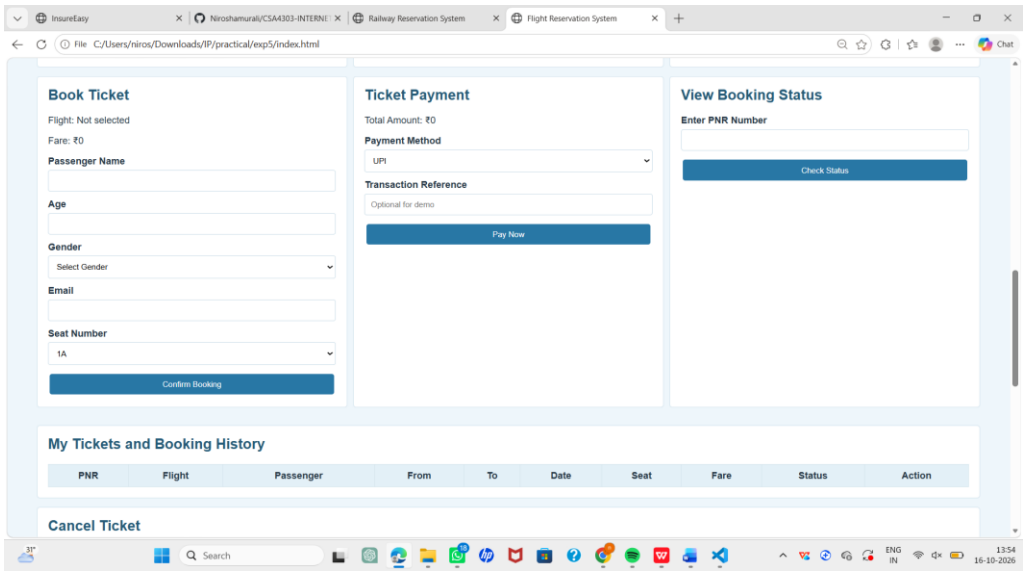
    function escapeHTML(value) {
        return value.replace(/[&<>'"]/g, function(character) {
            return {
                "&": "&amp;",
                "<": "&lt;",
                ">": "&gt;",
                '"': "&quot;",
                "'": "&#39;"
            }[character];
        });
    }
</script>

</body>
</html>
```

SAMPLE OUTPUT:







EXP 6

AIM :

To design and develop an Online Cake Ordering System using HTML, CSS, and JavaScript that allows users to search cakes, filter by category and price, customize cake orders, add cakes to the cart, enter shipping details, and view order confirmation.

CODE:

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <title>Sweet Treats - Cake Ordering</title>

  <style>
    * {
      box-sizing: border-box;
      margin: 0;
      padding: 0;
      font-family: Arial, sans-serif;
    }

    body {
      background: #fff7f8;
      color: #333;
    }

    header {
      background: #c2185b;
      color: white;
      padding: 22px;
```

```
    text-align: center;
}

header h1 {
    margin-bottom: 8px;
}

nav {
    background: #880e4f;
    padding: 12px;
    text-align: center;
}

nav a {
    color: white;
    text-decoration: none;
    margin: 0 15px;
    font-weight: bold;
}

.container {
    width: 92%;
    max-width: 1200px;
    margin: 25px auto;
}

h2 {
    color: #c2185b;
    margin-bottom: 18px;
}

.filters {
```

```
    display: flex;
    flex-wrap: wrap;
    gap: 12px;
    margin-bottom: 25px;
}

input, select, textarea, button {
    padding: 11px;
    border: 1px solid #ddd;
    border-radius: 6px;
    font-size: 15px;
}

.filters input, .filters select {
    flex: 1;
    min-width: 150px;
}

.cake-list {
    display: grid;
    grid-template-columns: repeat(auto-fit, minmax(230px, 1fr));
    gap: 20px;
}

.cake-card {
    background: white;
    border-radius: 12px;
    padding: 18px;
    text-align: center;
    box-shadow: 0 3px 12px #00000012;
}
```

```
.cake-image {
    font-size: 75px;
    background: #ffe3ec;
    padding: 15px;
    border-radius: 10px;
    margin-bottom: 12px;
}

.cake-card h3 {
    color: #880e4f;
    margin-bottom: 8px;
}

.cake-card p {
    margin: 8px 0;
}

.cake-card select,
.cake-card input {
    width: 100%;
    margin: 5px 0;
}

button {
    background: #c2185b;
    color: white;
    border: none;
    cursor: pointer;
    margin-top: 8px;
}

button:hover {
```

```
    background: #880e4f;
}

.section {
    background: white;
    padding: 22px;
    margin-top: 28px;
    border-radius: 12px;
    box-shadow: 0 3px 12px #0000000c;
}

.cart-item {
    padding: 12px 0;
    border-bottom: 1px solid #eee;
}

.cart-item button {
    margin-left: 8px;
}

.total {
    font-size: 20px;
    font-weight: bold;
    margin-top: 15px;
}

.shipping-form {
    display: grid;
    grid-template-columns: repeat(auto-fit, minmax(220px, 1fr));
    gap: 12px;
}
```

```
        .shipping-form textarea {
            grid-column: 1 / -1;
            min-height: 80px;
        }

        #confirmation {
            display: none;
            background: #e8f5e9;
            padding: 20px;
            border-radius: 10px;
            margin-top: 20px;
            overflow-wrap: anywhere;
        }

        footer {
            background: #880e4f;
            color: white;
            text-align: center;
            padding: 18px;
            margin-top: 35px;
        }

        @media (max-width: 600px) {
            nav a {
                display: inline-block;
                margin: 7px;
            }
        }
    </style>
</head>

<body>
```

```
<header>
  <h1>🎂 Sweet Treats</h1>
  <p>Fresh Cakes for Every Celebration!</p>
</header>

<nav>
  <a href="#cakes">Cakes</a>
  <a href="#cart">My Cart</a>
  <a href="#shipping">Checkout</a>
</nav>

<div class="container">

  <section id="cakes">
    <h2>Our Delicious Cakes</h2>

    <div class="filters">
      <input type="text" id="search"
          placeholder="Search cakes..."
          oninput="displayCakes()">

      <select id="category" onchange="displayCakes()">
        <option value="All">All Categories</option>
        <option value="Chocolate">Chocolate</option>
        <option value="Fruit">Fruit</option>
        <option value="Vanilla">Vanilla</option>
        <option value="Special">Special</option>
      </select>

      <select id="maxPrice" onchange="displayCakes()">
        <option value="10000">Any Price</option>
```

```
            <option value="400">Under ₹400</option>
            <option value="600">Under ₹600</option>
            <option value="800">Under ₹800</option>
        </select>
    </div>

    <div class="cake-list" id="cakeList"></div>
</section>

<section class="section" id="cart">
    <h2>🛒 Shopping Cart</h2>
    <div id="cartItems">Your cart is empty.</div>
    <p class="total">Total: ₹<span id="total">0</span></p>
</section>

<section class="section" id="shipping">
    <h2>📦 Shipping Details</h2>

    <form id="orderForm">
        <div class="shipping-form">
            <input type="text" id="customerName"
                placeholder="Full Name" required>

            <input type="email" id="email"
                placeholder="Email Address" required>

            <input type="tel" id="phone"
                placeholder="Phone Number"
                pattern="[0-9]{10}"
                maxlength="10"
                title="Enter a 10-digit phone number"
                required>
```

```
            <input type="text" id="pincode"
                placeholder="6-digit PIN Code"
                pattern="[0-9]{6}"
                maxlength="6"
                title="Enter a 6-digit PIN code"
                required>

            <textarea id="address"
                placeholder="Complete Shipping Address"
                required></textarea>

            <input type="date" id="deliveryDate"
                aria-label="Preferred delivery date"
                required>
        </div>

        <button type="submit">Place Order</button>
    </form>

    <div id="confirmation" aria-live="polite"></div>
  </section>

</div>

<footer>
  <p>© 2026 Sweet Treats Cake Shop | Made with ♥</p>
</footer>

<script>
  const cakes = [
    { id: 1, name: "Chocolate Cake", category: "Chocolate",
```

```
    price: 450, emoji: "🍫" },
  { id: 2, name: "Black Forest Cake", category: "Chocolate",
    price: 550, emoji: "🍰" },
  { id: 3, name: "Strawberry Cake", category: "Fruit",
    price: 400, emoji: "🍓" },
  { id: 4, name: "Vanilla Cake", category: "Vanilla",
    price: 350, emoji: "🎂" },
  { id: 5, name: "Pineapple Cake", category: "Fruit",
    price: 420, emoji: "🍍" },
  { id: 6, name: "Red Velvet Cake", category: "Special",
    price: 650, emoji: "❤️" }
];

let cart = [];

// Display cakes with search and filters
function displayCakes() {
  const search = document.getElementById("search")
    .value.toLowerCase();
  const category = document.getElementById("category").value;
  const maxPrice = Number(
    document.getElementById("maxPrice").value
  );

  const filtered = cakes.filter(cake =>
    cake.name.toLowerCase().includes(search) &&
    (category === "All" || cake.category === category) &&
    cake.price < maxPrice
  );

  const cakeList = document.getElementById("cakeList");
```

```
if (filtered.length === 0) {
    cakeList.innerHTML = "<p>No cakes found.</p>";
    return;
}

cakeList.innerHTML = filtered.map(cake => `
    <div class="cake-card">
        <div class="cake-image">${cake.emoji}</div>
        <h3>${cake.name}</h3>
        <p>Category: ${cake.category}</p>
        <p><strong>₹${cake.price}</strong></p>

        <label>Size:</label>
        <select id="size-${cake.id}">
            <option value="0.5">0.5 kg</option>
            <option value="1" selected>1 kg</option>
            <option value="2">2 kg</option>
        </select>

        <label>Flavor:</label>
        <select id="flavor-${cake.id}">
            <option>Original</option>
            <option>Extra Chocolate</option>
            <option>Strawberry</option>
            <option>Vanilla</option>
        </select>

        <input type="text" id="message-${cake.id}"
               placeholder="Cake message (optional)"
               maxlength="60">

        <button type="button"
```

```
                onclick="addToCart(${cake.id})">
            Add to Cart
          </button>
        </div>
    `).join("");
}

// Add customized cake to cart
function addToCart(id) {
    const cake = cakes.find(item => item.id === id);
    const size = Number(
        document.getElementById(`size-${id}`).value
    );
    const flavor = document.getElementById(`flavor-${id}`).value;
    const message = document.getElementById(`message-${id}`)
        .value.trim();

    const item = {
        cartId: Date.now() + Math.random(),
        name: cake.name,
        size: size,
        flavor: flavor,
        message: message,
        price: Math.round(cake.price * size)
    };

    cart.push(item);
    displayCart();
    alert(cake.name + " added to your cart!");
}

// Display cart and total price
```

```
function displayCart() {
    const cartItems = document.getElementById("cartItems");

    if (cart.length === 0) {
        cartItems.textContent = "Your cart is empty.";
    } else {
        cartItems.innerHTML = cart.map(item => `
            <div class="cart-item">
                <strong>${escapeHTML(item.name)}</strong>
                (${item.size} kg)
                <br>Flavor: ${escapeHTML(item.flavor)}
                <br>Message: ${escapeHTML(item.message || "None")}
                <br>Price: ₹${item.price}
                <button type="button"
                    onclick="removeFromCart('${item.cartId}')">
                    Remove
                </button>
            </div>
        `).join("");
    }

    const total = cart.reduce((sum, item) => sum + item.price, 0);
    document.getElementById("total").textContent = total;
}

function removeFromCart(cartId) {
    cart = cart.filter(item =>
        String(item.cartId) !== String(cartId)
    );
    displayCart();
}
```

```
// Prevent HTML injection in displayed user-entered text
function escapeHTML(value) {
    return String(value).replace(/[&<>"']/g, char => ({
        "&": "&amp;",
        "<": "&lt;",
        ">": "&gt;",
        "": "&quot;",
        "'": "&#39;"
    })[char]);
}

// Place order and show confirmation
document.getElementById("orderForm").addEventListener(
    "submit", function(event) {
        event.preventDefault();

        if (cart.length === 0) {
            alert("Please add at least one cake to your cart.");
            return;
        }

        const name = document.getElementById("customerName").value.trim();
        const email = document.getElementById("email").value.trim();
        const phone = document.getElementById("phone").value.trim();
        const pincode = document.getElementById("pincode").value.trim();
        const address = document.getElementById("address").value.trim();
        const deliveryDate = document.getElementById("deliveryDate").value;

        if (!name || !email || !address ||
            !/^[0-9]{10}$/.test(phone) ||
            !/^[0-9]{6}$/.test(pincode)) {
            alert("Please enter valid shipping details.");
```

```
    return;
}

const selectedDate = new Date(deliveryDate + "T00:00:00");
const today = new Date();
today.setHours(0, 0, 0, 0);

if (!deliveryDate || selectedDate < today) {
    alert("Please select today or a future delivery date.");
    return;
}

const orderId = "CAKE" + Date.now().toString().slice(-8);
const total = cart.reduce((sum, item) => sum + item.price, 0);

const confirmation = document.getElementById("confirmation");
confirmation.style.display = "block";
confirmation.innerHTML = `
    <h2>🎉 Order Confirmed!</h2>
    <p><strong>Order ID:</strong> ${orderId}</p>
    <p><strong>Customer:</strong> ${escapeHTML(name)}</p>
    <p><strong>Email:</strong> ${escapeHTML(email)}</p>
    <p><strong>Phone:</strong> ${escapeHTML(phone)}</p>
    <p><strong>Delivery Address:</strong>
      ${escapeHTML(address)}, ${escapeHTML(pincode)}</p>
    <p><strong>Delivery Date:</strong>
      ${escapeHTML(deliveryDate)}</p>
    <p><strong>Total Amount:</strong> ₹${total}</p>
    <p>Your demo order has been placed successfully!</p>
    <button type="button" onclick="window.print()">
       Print Confirmation
    </button>
```

```
            `;

            confirmation.scrollIntoView({ behavior: "smooth" });
        }
    );

    // Set the earliest delivery date to today
    const dateInput = document.getElementById("deliveryDate");
    const todayString = new Date();
    const localToday = [
        todayString.getFullYear(),
        String(todayString.getMonth() + 1).padStart(2, "0"),
        String(todayString.getDate()).padStart(2, "0")
    ].join("-");

    dateInput.min = localToday;

    displayCakes();
</script>

</body>
</html>
```

SAMPLE OUTPUT:

